

Phase-based video motion magnification with handheld cameras

Zongdi Zang, Xuezhi Yang^{*}, Gang Zhang, Qin Xu

School of Computer Science and Information Engineering, Hefei University of Technology, 193 Tunxi Road, Hefei, 230009, Anhui, China
Anhui Key Laboratory of Industry Safety and Emergency Technology, Hefei University of Technology, 420 Feicui Road, Hefei, 230009, Anhui, China

ARTICLE INFO

Communicated by Xiaobai Li

Dataset link: <https://github.com/singleThreadD/PVMHC>

Keywords:

Eulerian motion magnification
Phase-based
Handheld cameras
Sparse coding
Orthogonal matching pursuit

ABSTRACT

Eulerian video magnification enables the visualization of subtle motions in videos that are otherwise imperceptible to the human eye. However, when the imaging device moves, the magnified video often suffers from distortion. To address this issue, this paper proposes a motion magnification method tailored for videos captured with handheld cameras. To achieve robust magnification, the method employs phase-based motion representation and introduces acceleration-domain filtering to suppress large-scale camera movements. Specifically, based on the sparsity of subtle motions in the frequency domain and their localized nature in the spatial domain, a region-guided orthogonal matching pursuit (RGOMP) algorithm is proposed to separate random handheld camera motion from regular target subtle motion. RGOMP maps the full-field phase sequences onto a carefully designed motion dictionary and assigns different weights to the sequences according to their spatial locations, enabling the computation of matching coefficients. The reconstructed phase signals are then multiplied by a magnification factor and superimposed onto the original video, thereby achieving the magnification of subtle motions. Experimental results show that, compared with existing state-of-the-art methods, the proposed approach more effectively magnifies subtle motion signals in handheld-camera videos and demonstrates superior performance.

1. Introduction

The visualization of dynamic phenomena in videos plays a crucial role in scientific analysis, medical diagnosis, and engineering monitoring (Wang and Zhu, 2023; Thomas et al., 2017). Within this framework, the enhancement of subtle motions, which are imperceptible to the naked eye yet rich in physical and physiological information, has emerged as a key research direction. Subtle motions are widely present in various natural phenomena, such as the gentle sway of tall structures, the slight vibrations during mechanical operation, the rhythmic pulsation of the human heartbeat (Zhao et al., 2022; Shao et al., 2023), or the micro-expressions of the face (W. Cai et al., 2024). These motions convey valuable information about the physical properties and functions of objects. However, these subtle yet meaningful movements are often difficult for the human eye to perceive.

To enhance subtle motions in videos, Wu et al. (2012) proposed the linear Eulerian video magnification to amplify specific motion frequency bands through intensity changes using Eulerian fluid dynamics. Compared to motion magnification methods based on the Lagrangian perspective (Liu et al., 2005; Elgharib et al., 2015), the Eulerian approach offers advantages in global motion extraction and

computational complexity. Wadhwa et al. (2013) proposed the phase-based Eulerian video magnification method (Phase-based), which represents motion signals through local phase variations, achieving higher magnification factors and reduced motion magnification artifacts. Eulerian video magnification has found widespread applications in various scientific research areas, such as structural modal analysis (Chen et al., 2015; Davis et al., 2015; Yang et al., 2017; Poozesh et al., 2017; Sarrafi et al., 2018), health monitoring of ancient buildings (Fioriti et al., 2018, 2022), biomedical applications (McLeod et al., 2014; Al-Naji and Chahl, 2016; Shabi et al., 2020), and micro-expression recognition (Wang et al., 2017; Ngo et al., 2018).

There is ongoing research aimed at enhancing the performance of Eulerian video magnification. Wadhwa et al. (2014) addressed the computational cost of Phase-based by utilizing the Riesz transform to extract spatial local phase. Wu et al. (2018) applied the principal component analysis to improve the noise resistance of Phase-based. Tae-Hyun Oh et al. (2018) employed deep convolutional neural networks to learn spatial sub-band representations, replacing Laplacian pyramids (Wu et al., 2012) and complex steerable pyramids (Wadhwa et al., 2013). This, combined with temporal filtering, yielded promising magnification results. Takeda (Takeda et al., 2019) utilized the fractional anisotropy of natural subtle motions in the temporal domain to

^{*} Corresponding author.

E-mail addresses: 2024820021@hfut.edu.cn (Z. Zang), xzyang@hfut.edu.cn (X. Yang), 2018180099@mail.hfut.edu.cn (G. Zhang), 2023010056@mail.hfut.edu.cn (Q. Xu).

<https://doi.org/10.1016/j.cviu.2026.104806>

Received 26 May 2025; Received in revised form 26 March 2026; Accepted 7 May 2026

Available online 13 May 2026

1077-3142/© 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

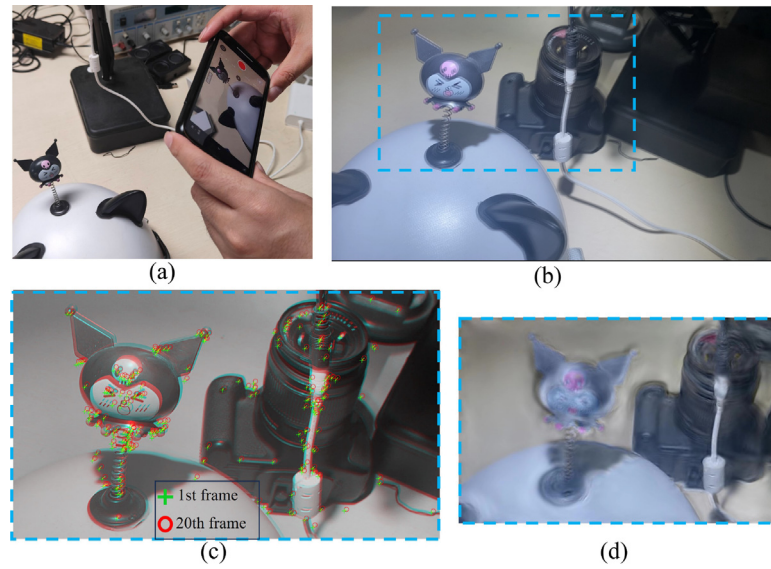


Fig. 1. Unstable footage caused by handheld camera shooting. (a) Recording a video with a mobile phone. (b) The snapshot of the recorded video. (c) The corner points in the 1st frame and the 20th frame, along with the transformation vector between these two frames. (d) The magnification results with the phase-based video motion magnification (Phase-based) (Wadhwa et al., 2013). The inter-frame motion vectors indicate that the irregular shaking of the hand causes overall motion in the image. Directly using Phase-based would amplify these motion interference, thus failing to capture the subtle motion signals inherent in the video.

separate meaningful subtle motions from irrelevant imaging noise in videos. In recent years, research on video magnification has focused on using deep neural networks to reveal the relational characteristics between the original video and the magnified video, thereby constructing a mapping model for magnification visualization. This approach has now been extended from 2D to 3D (Feng et al., 2023) and incorporates frequency domain features with spatial domain features (Singh et al., 2023). On the other hand, some studies have improved the quality of motion magnification by combining the Eulerian and Lagrangian perspectives (Chen et al., 2023). More recently, Lado-Roigé and Pérez (Lado-Roigé and Pérez, 2023) proposed a Swin Transformer-based video motion magnification framework (STB-VMM), which leverages hierarchical self-attention to capture long-range spatiotemporal dependencies, further demonstrating the effectiveness of transformer architectures in video motion magnification tasks.

The aforementioned studies, represented by Chen et al. (2023), focus on enhancing the performance of video motion magnification but are all based on the assumption of a stationary or fixed-position camera. When using a handheld camera, i.e., when there is camera motion, these methods may fail. Since the camera is at the observation point, even a slight shake of the camera can cause a significant shift in the captured image, severely affecting the accuracy of subtle motion extraction. In the field of video magnification, researchers have begun to address the problem of separating large movements from subtle motions. Elgharib et al. (2015) proposed a method for motion magnification in the presence of large motions by tracking and isolating manually selected regions to eliminate the large motion signals, thus extracting and amplifying subtle motions within the target region. But if there are significant variations in the grayscale values of the target at different positions in the image, or if the background texture is complex, this method may fail. On the other hand, Zhang et al. (2017) introduced a new technique called video acceleration magnification. This approach is purely Eulerian and the proposed acceleration filter effectively distinguishes between linear large motion and subtle motion signals with acceleration components. Subsequently, Takeda et al. (2018, 2022) improved the existing acceleration magnification framework, making it applicable in the presence of fast, nonlinear large motions.

Compared to the large movements of the object itself, camera motion exhibits greater randomness and has a global spatial effect, making the acceleration magnification method not directly applicable

to handheld cameras, as shown in Fig. 1. In this case, random motion signals are dispersed across the entire video frame, posing challenges in isolating subtle motions of interest. Alam et al. (2017) estimated the respiratory rate of cats in videos recorded with handheld cameras. To achieve this, they employed motion magnification with a basis in video stabilization (Wang et al., 2024; Zhang et al., 2015; Kim et al., 2014). The process of video stabilization entails tracking feature points in each frame to estimate the global motion matrix. This matrix is then used to eliminate the global motion signal, resulting in a stabilized video. Following stabilization, phase-based motion magnification was applied to these videos. However, it is worth noting that while video stabilization effectively eliminates camera motion, it may also inadvertently suppress subtle motions of the target, potentially impacting the quality of video magnification. In recent years, E. Cai et al. (2024) investigated a vision-based automatic structural motion estimation method for scenarios involving 3D camera motion effects. Their approach explicitly estimates the motion components introduced by the camera and compensates for the global camera motion, enabling structural motion estimation under nonstationary camera conditions. Nevertheless, the effectiveness of such camera motion estimation strategies may depend on the availability of sufficient and reliable feature points in the background regions of the video, and their performance could be affected when the background texture is weak or feature distribution is limited.

This paper proposes a novel framework for subtle motion magnification in videos captured by handheld cameras, where camera-induced motion and target micro-motion are intrinsically coupled. Instead of relying on explicit camera motion estimation or stabilization, we formulate the problem as a motion separation task driven by the distinct spatial and temporal characteristics of different motion components. Specifically, we observe that camera motion typically exhibits globally distributed, high-amplitude variations, while subtle target motions are spatially localized and relatively weak, yet exhibit structured sparsity in the frequency domain. Based on this observation, we introduce a two-stage motion disentanglement strategy. First, a jerk-aware acceleration filter is employed to suppress dominant global motion components in the phase domain. Unlike its conventional use for large motion analysis, here it serves as a coarse separation mechanism tailored to handheld camera scenarios. Second, to further isolate subtle motions from residual interference, we propose a Region-Guided Orthogonal Matching Pursuit (RGOMP) framework. Rather than performing standard sparse

reconstruction or simple region-based weighting, RGOMP incorporates region priors as guidance to adaptively select phase components that are most consistent with the underlying micro-motion. This results in a structured sparse representation that aligns with the physical characteristics of subtle motion, enabling effective suppression of spatially inconsistent camera-induced variations. Finally, the refined phase signals are magnified and reconstructed using a steerable pyramid to generate videos with enhanced subtle motions. By explicitly leveraging motion-specific priors and structured sparsity, the proposed framework provides a new perspective for motion magnification under complex observation conditions, beyond conventional stabilization-based or purely filtering-based approaches.

This paper is organized as follows. Section 1 introduces the research background and motivation. Section 2 explains the main theoretical problems addressed in this study. Section 3 presents the proposed method. Section 4 provides a detailed description of the experiments, including the results and analysis. Section 5 summarizes the research content, significance, and limitations of this study.

2. Problem statement

Let $I(x, t)$ represent the one-dimensional image intensity signal at time t , and let $\delta(x, t)$ represent the local motion between time 0 and t , then there is $I(x, 0) = f(x)$, $I(x, t) = f(x + \delta(x, t))$. The purpose of video motion magnification is to synthesize such a signal based on the magnification factor α : $\hat{I}(x, t) = f(x + (1 + \alpha)\delta(x, t))$. Phase-based video magnification (Wadhwa et al., 2013) characterizes local motion through changes in the local spatial phase and then magnifies subtle motions by manipulating the phase. To achieve this, $f(x + \delta(x, t))$ is expanded into a Fourier series

$$f(x + \delta(x, t)) = \sum_{\omega=-\infty}^{\infty} S_{\omega}(x, t) = \sum_{\omega=-\infty}^{\infty} A_{\omega} e^{j\omega(x + \delta(x, t))} \quad (1)$$

where $S_{\omega}(x, t)$ represents the subband with a spatial frequency of ω . To obtain the subtle motion within the frequency band of interest, a temporal band-pass filter is applied to the phase $\omega(x + \delta(x, t))$, resulting in $B_{\omega}(x, t)$. For simplification of the analysis, assuming that $\delta(x, t)$ lies within the specified frequency band, the filtering process is equivalent to removing only the DC component, resulting in $B_{\omega}(x, t) = \omega\delta(x, t)$. The filtered phase $B_{\omega}(x, t)$ is multiplied by α and then added to the subband $S_{\omega}(x, t)$, resulting in the amplified subband

$$\hat{S}_{\omega}(x, t) = S_{\omega}(x, t) e^{j\alpha B_{\omega}} = A_{\omega} e^{j\omega(x + (1 + \alpha)\delta(x, t))} \quad (2)$$

The amplified subband $\hat{S}_{\omega}(x, t)$ contains a displacement that is $(1 + \alpha)$ times the input. By combining each amplified spatial subband, a motion-magnified video sequence $f(x + (1 + \alpha)\delta(x, t))$ can be obtained. Notably, the magnification factor α has an upper limit due to the video's limited spatial support. For an octave bandwidth steerable pyramid, the boundary for α is $\alpha < \lambda/(4\delta(x, t))$, where $\lambda = 1/\omega$ denotes the spatial wavelength.

When recording video with a handheld camera, it is challenging to maintain stability, resulting in “camera motion” in the video, or in other words, false motion caused by changes in the camera's position. Thus, the intensity signal at time t in Eq. (1) should be rewritten as $I'(x, t) = f(x + \delta'(x, t))$, $\delta'(x, t) = \delta_C(x, t) + \delta_T(x, t)$. Where $\delta_C(t)$ and $\delta_T(t)$ represent the camera motion and the subtle motion, respectively. According to the phase-based video magnification, the final magnified video can be expressed as

$$\begin{aligned} \hat{I}'(x, t) &= f(x + (1 + \alpha)\delta'(x, t)) \\ &= f(x + (1 + \alpha)(\delta_C(x, t) + \delta_T(x, t))) \end{aligned} \quad (3)$$

The indiscriminate amplification of these two types of motion results in chaotic motion signals in the output video, making it difficult to identify the true movement of the target object. Additionally, because the camera motion $\delta_C(t)$ typically has a large amplitude, the allowable magnification factor in Eq. (3) is likely to exceed the upper limit,

leading to artifacts in the output video. The magnified video we expect should be

$$\hat{I}'(x, t) = f(x + \delta_C(x, t) + (1 + \alpha)\delta_T(x, t)) \quad (4)$$

Therefore, accurately separating the camera motion $\delta_C(x, t)$ and the subtle motion $\delta_T(x, t)$ contained in $\delta'(x, t)$ is a key issue for achieving subtle motion magnification under camera instability. In the phase-based framework, this problem is equivalent to separating the target subtle motion phase $\omega\delta_T(t)$ and the camera motion phase $\omega\delta_C(t)$ from the mixed phase signal $\omega\delta'(t)$.

3. Proposed method

The proposed method can be divided into four parts: Spatial Phase Extraction, Temporal Acceleration Filtering, Phase Restoration by RGOMP, and Subtle Motion Magnification, as shown in Fig. 2.

3.1. Spatial phase extraction

Due to the robustness of motion representation in the phase domain compared to intensity, and its higher magnification factors with fewer artifacts (Fleet and Jepson, 1990; Gautama and Van Hulle, 2002), we extract subtle motion signals through the local phase variations of pixels. The local phase needs to be calculated from the intensity matrix of the image. The source video is converted from the RGB to the YIQ color space, and the intensity signal of the video is obtained by extracting the Y-channel image sequence $Y(x, y, t)$. Based on the intensity signal $Y(t_0)$ of the video frame at time t_0 , the local phase of pixels can be extracted using a complex steerable filter (Freeman and Adelson, 1991) in the frequency domain. This can further be used to construct a complex steerable pyramid (CSP), enabling multi-scale and multi-orientational video magnification.

Specifically, the intensity signal $Y(x, y, t_0)$ is transformed to the frequency domain through a Fourier transform to obtain $\hat{Y}(u, v, t_0)$, and then it is filtered using a complex steerable filter $\Psi_{\omega, \theta}$ with scale ω and orientation θ

$$R_{\omega, \theta}(x, y, t_0) = \mathcal{F}^{-1}(\hat{Y}(u, v, t_0) \odot \Psi_{\omega, \theta}) \quad (5)$$

where $\odot$ represents the Hadamard product, $\mathcal{F}^{-1}$ represents the inverse Fourier transform. Typically, the subband response $R_{\omega, \theta}(x, y, t_0)$ at a specific scale ω and a specific orientation θ is defined as a “layer” of the steerable pyramid, denoted as $R^l(x, y, t_0)$. The phase map of the subband response $R^l(x, y, t_0)$ for the image at frame t_0 is given by the argument of it

$$\phi^l(x, y, t_0) = \arg(R^l(x, y, t_0)) \quad (6)$$

By processing each frame of the image, a three-dimensional phase tensor $\phi^l(x, y, t)$ can be obtained, abbreviated as $\phi^l(x, t)$.

3.2. Temporal acceleration filtering

The camera motion induced by handheld cameras typically exhibits linear characteristics, while the intended subtle movements of the target possess large acceleration components. Therefore, temporal acceleration filtering can be employed to preliminarily suppress camera motion. To prevent the interference of inevitable large non-linear components in camera motion from affecting the extraction of subtle movements, we introduced the Jerk-aware acceleration filter (Takeda et al., 2018).

The Jerk-aware filter comprises two components: the ordinary acceleration kernel and the jerk kernel, which are designed to filter out slow large motions and fast large motions, respectively. In the video domain, camera motion, compared to the subtle changes of interest, can clearly be defined as large motion. Therefore, the Jerk-aware filter is an ideal solution. The ordinary acceleration kernel $H_A(t)$ consists of a second-order derivative operation and a Gaussian function to prevent spurious

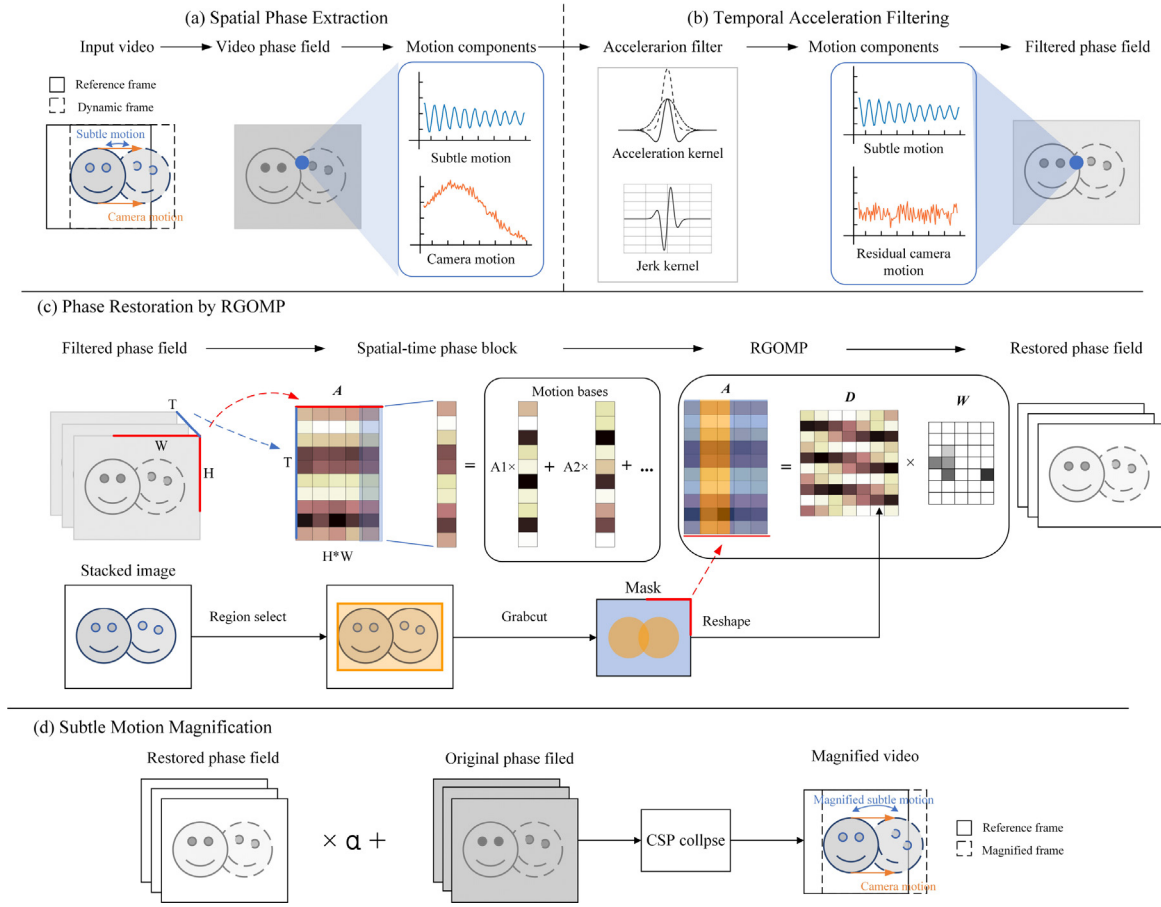


Fig. 2. The proposed method.

resolution. The phase signal after acceleration filtering ($\phi_{Acc_\sigma}^l(\mathbf{x}, t)$) can be expressed as

$$\begin{aligned}\phi_{Acc_\sigma}^l(\mathbf{x}, t) &= \phi^l(\mathbf{x}, t) \otimes \frac{\partial^2}{\partial t^2} G_\sigma(t) \\ &= \frac{\partial^2}{\partial t^2} \phi^l(\mathbf{x}, t) \otimes G_\sigma(t)\end{aligned}\quad (7)$$

where $G_\sigma(t)$ represents the standard Gaussian function. Based on the given frequency of interest (f_i), let $\sigma = f_s/(4\sqrt{2}f_i)$ to select the acceleration signal corresponding to the specified frequency.

The main purpose of the jerk filter kernel is to remove jerk signals at a specified frequency. The phase signal after being filtered by the original jerk kernel ($\phi_{Jerk_\sigma}^l(\mathbf{x}, t)$) can be expressed as

$$\begin{aligned}\phi_{Jerk_\sigma}^l(\mathbf{x}, t) &= \phi^l(\mathbf{x}, t) \otimes \frac{\partial^3}{\partial t^3} G_\sigma(t) \\ &= \frac{\partial^3}{\partial t^3} \phi^l(\mathbf{x}, t) \otimes G_\sigma(t)\end{aligned}\quad (8)$$

Since jerk is ultimately used for signal suppression, it is necessary to construct a smoothing term using it, which is expressed as

$$\begin{aligned}\phi_{nJerk_\sigma}^l(\mathbf{x}, t) &= \frac{|\phi_{Jerk_\sigma}^l(\mathbf{x}, t)| - \min_{\mathbf{x}, t} |\phi_{Jerk_\sigma}^l(\mathbf{x}, t)|}{\max_{\mathbf{x}, t} |\phi_{Jerk_\sigma}^l(\mathbf{x}, t)| - \min_{\mathbf{x}, t} |\phi_{Jerk_\sigma}^l(\mathbf{x}, t)|}\end{aligned}\quad (9)$$

$$JAF_\sigma^l(\mathbf{x}, t) = (1 - \phi_{nJerk_\sigma}^l(\mathbf{x}, t))^\beta \quad (10)$$

where β is the coefficient used to control the degree of smoothing. However, within the steerable pyramid framework, the spatial scales of the signals at each level are different, and the large motions they contain also vary. Therefore, the jerk-based smoothing must reflect the

differences between layers. Based on this, the jerk-based smoothing term is adjusted as follows

$$JAF_{\sigma, \lambda}^l(\mathbf{x}, t) = JAF_\sigma^l(\mathbf{x}, t)^{1/\lambda} \quad (11)$$

where λ is the downsampling factor of the steerable pyramid at each level. On the other hand, the lower pyramid levels contain more subtle motions and are less sensitive to large movements, so it is necessary to propagate motion information from higher levels to the motion signals of lower levels, as expressed by

$$pJAF_{\sigma, \lambda}^l(\mathbf{x}, t) = \prod_{i=l}^{l+N} \mathcal{R}(JAF_{\sigma, \lambda}^i(\mathbf{x}, t)) \quad (12)$$

where N is the number of pyramid levels for this correction, and the function $\mathcal{R}(\cdot)$ adjusts the filter size from pyramid level i to l . Finally, applying the corrected jerk-based smoothing term to the result of the ordinary acceleration filter enables the preservation of subtle motions while suppressing the overall large motions generated by the camera

$$\phi_f^l(\mathbf{x}, t) = \phi_{Acc_\sigma}^l(\mathbf{x}, t) \times pJAF_{\sigma, \lambda}^l(\mathbf{x}, t) \quad (13)$$

3.3. Phase restoration by RGOMP

The Temporal Acceleration Filtering process eliminates large-scale camera motion interference, but small-scale camera motion may still remain in the phase signals. In video magnification tasks, since the target subtle motions are usually periodic reciprocating motions, they exhibit sparsity in the frequency domain. In contrast, camera motions are relatively random. Leveraging this characteristic, we employ sparse coding to extract relatively regular subtle motions.

Reducing the dimensionality of the globally filtered phase signal $\phi_f^l(x, y, t)$ after acceleration filtering, preserving the temporal dimension t , and arranging all spatial coordinates (x, y) into a single dimension, we can obtain subtle motion matrix $\mathbf{A}(t, u)$, $t = 1, 2, \dots, T$, $u = 1, 2, \dots, H \times W$, where H and W are the height and width of the video frame, respectively. Normally, the raw subtle motion (SM) matrix ($\mathbf{A}$) contains an ideal SM matrix ($\mathbf{A}_{\text{subtle}}$) and a motion interference matrix ($\mathbf{A}_{\text{camera}}$), denoted as $\mathbf{A} = \mathbf{A}_{\text{subtle}} + \mathbf{A}_{\text{camera}}$. The objective of the RGOMP algorithm is to construct a synthetic SM matrix $\mathbf{S} = [s^1, s^2, \dots, s^{H \times W}]$.

A discrete signal can be represented as a linear combination of several sinusoidal bases with different amplitudes, frequencies, and phases. The n th row of the synthesized SM matrix ($\mathbf{S}$), represented as s^n , serves as an approximation of the ideal SM signals and can be composed of sinusoidal bases with frequencies tightly constrained around the target frequency. Based on this, a motion dictionary ($\mathbf{D}$) is constructed to store the aforementioned sinusoidal bases as follows

$$\begin{aligned} D_{ij} &= \sin(2\pi k_i t / F_s + \rho_j) \\ \mathbf{D} &= [D_{11}, D_{12}, \dots, D_{1j}, \dots, D_{1J}] \end{aligned} \quad (14)$$

where D_{ij} denotes a sinusoidal basis characterized by a frequency k_i and an initial temporal phase ρ_j . The parameters k_i and ρ_j correspond to the i th frequency scale and the j th phase scale, respectively. The frequency scales k_i are separated by intervals of 1/100 Hz, while the phase scales ρ_j are spaced at intervals of $\pi/10$. t represents the sequence of video frames, and F_s indicates the video's frame rate. Using this motion dictionary, the synthesized SM matrix ($\mathbf{S}$) can be formulated as follows

$$\mathbf{S} = \mathbf{D} \cdot \mathbf{W} \quad (15)$$

where $\mathbf{W}$ represents the matching coefficient matrix.

Considering the sparsity of the subtle motion signal to be magnified in the frequency domain, the raw SM signal at any position ($\mathbf{a}^u$) can be composed of only a small number of sinusoidal bases. In other words, the coefficient matrix $\mathbf{W}$ is sparse, containing only a small number of nonzero elements. The sparsity of the coefficient matrix can be modeled by minimizing the ℓ_0 -norm, i.e., $\min \|\mathbf{W}\|_0$. On the other hand, to ensure the fidelity of the subtle motion, the ℓ_2 -norm of the difference between the original SM matrix and the synthetic SM matrix is introduced, i.e., $\min \|\mathbf{A} - \mathbf{S}\|_2$. Building on the above analysis, the construction of the ideal SM matrix can be formulated as the following optimization problem

$$\min_{\mathbf{W}} \|\mathbf{A} - \mathbf{D}\mathbf{W}\|_2 + \mu \|\mathbf{W}\|_0 \quad (16)$$

In addition to sparsity in the frequency domain, the target subtle motion also exhibits a certain spatial regularity in the task of motion magnification using a handheld camera. Specifically, the motion signals are concentrated within a fixed region, while camera motion is distributed across the entire image plane. In other words, the contribution of the signals $\mathbf{a}^u$ at different spatial locations in the original subtle motion matrix ($\mathbf{A}$) to the recovery of the true subtle motion is different. To address this issue, we manually select regions on the video image to guide the generation of a spatial domain template $\mathbf{M}$, which is used to control the weight of signals at different positions for the recovery of subtle motion. The manually selected region is also referred to as the ROS (Region of Selected).

To construct the template $\mathbf{M}$, we first preprocess the video to create a "stacked image"

$$\mathbf{RGB}_{\text{stacked}} = \frac{1}{T} \sum_{t=1}^T \mathbf{RGB}(t) \quad (17)$$

where $\mathbf{RGB}(t)$ represents the RGB signal of the video at time t . In the case of camera motion, the position of the target object undergoing subtle motion in the frame is uncertain. Therefore, it is necessary to stack images from all moments to ensure that the position information of the target object is not missed. The user is then asked to select a

rectangular ROS containing the subtle motion target. The GrabCut algorithm (Rother et al., 2004) is applied to process the $\mathbf{RGB}_{\text{stacked}}$, iteratively segmenting the target object from the background and generating the segmentation template $\mathbf{\Gamma}^o$

$$\mathbf{\Gamma}^o = \mathcal{T}(\mathbf{RGB}_{\text{stacked}}) \quad (18)$$

where $\mathcal{T}$ represents the operation of GrabCut. $\mathbf{\Gamma}^o$ has a truncation characteristic, taking the value of 1 in the ROS and 0 in the background region, and cannot be directly applied to the original SM matrix ($\mathbf{A}$). The reason is that the phase of subtle motion has a "diffusion effect" in space, as shown in Fig. 3. To address this, all the 0 values in $\mathbf{\Gamma}^o$ are set to η ($0 < \eta < 1$), i.e.

$$\mathbf{\Gamma}(y, x) = \begin{cases} 1, & \mathbf{\Gamma}^o(y, x) = 1 \\ \eta, & \mathbf{\Gamma}^o(y, x) = 0 \end{cases} \quad (19)$$

where η is also referred to as the background retention coefficient. Then, $\mathbf{\Gamma}$ needs to be reshaped and copied in each frame to match the size of the original sub-SM matrix ($\mathbf{A}$), thereby obtaining the mask $\mathbf{M}$

$$\mathbf{M}(t, (x-1) * H + y) = \mathbf{\Gamma}(y, x), \quad t = 1, 2, \dots, T \quad (20)$$

After obtaining the weight matrix ($\mathbf{M}$), the optimization problem described by Eq. (16) is transformed into

$$\min_{\mathbf{W}} \|\mathbf{M} \odot (\mathbf{A} - \mathbf{D}\mathbf{W})\|_2 + \mu \|\mathbf{W}\|_0 \quad (21)$$

where $\odot$ denotes the Hadamard product. Eq. (21) is an NP-hard problem but can be solved using a greedy algorithm, RGOMP. To solve the weighted sparse reconstruction problem in Eq. (21), a greedy strategy based on the region-guided orthogonal matching pursuit (RGOMP) is adopted. For a given weighted sub-SM signal $\mathbf{m}^1 \odot \mathbf{a}^1$, the algorithm iteratively selects sinusoidal bases from the motion dictionary that exhibit the highest correlation with the current residual signal. Specifically, the similarity between each sinusoidal basis and the masked sub-SM is evaluated using their inner product, and the basis achieving the maximum correlation is selected as the optimal matching atom.

Once the optimal basis is determined, the corresponding matching coefficient is estimated via least-squares projection, and the residual signal is updated by subtracting the contribution of the selected basis. The matching error is then computed based on the energy of the weighted residual signal and is used as the stopping criterion. This iterative procedure continues until the matching error falls below a predefined threshold or the maximum number of iterations is reached.

After convergence, the synthesized sub-SM signal s^1 is represented as a linear combination of a small number of selected sinusoidal bases and their associated coefficients. The same reconstruction process is independently applied to all spatial locations in the subtle motion matrix. By assembling the recovered sub-SMs and their coefficients into the corresponding positions, the synthesized subtle motion matrix $\mathbf{S}$ is finally obtained according to Eq. (15).

Since the physical meaning of the synthesized SM signal is still phase, $\mathbf{S}$ is denoted as $\phi_{\text{res}}^l(x, t)$. Fig. 4 illustrates the spatial distribution of phase difference signals at several key steps of the proposed method. For the original phase signal, the phase difference caused by the subtle motion of the target object (the exciter) and the phase difference from the camera motion are spatially mixed, appearing as a whole. After acceleration filtering, the outline of the exciter becomes distinguishable in the phase distribution. Following sparse reconstruction of the phase, the spatial distribution map reveals only the phase variations caused by the exciter's vibrating plate.

3.4. Subtle motion magnification

By using the complex steerable pyramid (CSP), after magnifying the subtle motion phase at each layer, it is added to the original phase signal

$$\phi_{\text{Mag}}^l(\mathbf{x}, t) = \phi^l(\mathbf{x}, t) + \alpha \phi_{\text{res}}^l(\mathbf{x}, t) \quad (22)$$

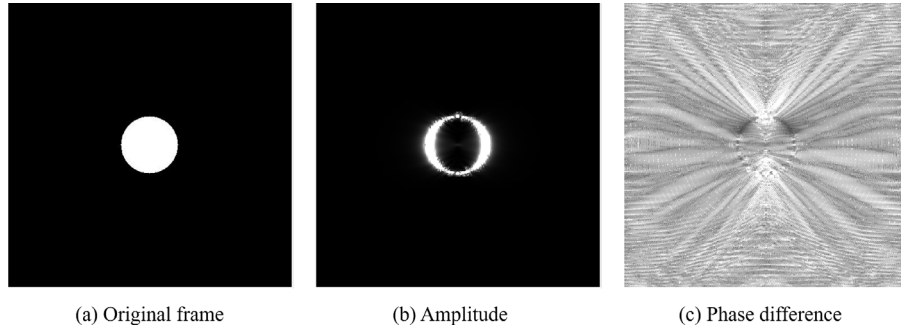


Fig. 3. The diffusion effect of the phase of subtle motion in space. (a) The original frame of a simulated vibrating ball video. (b) The amplitude obtained by processing the original frame through a steerable filter. (c) The phase difference between the original video frame and the frame 10 frames later, with the phase signal processed through a steerable filter. The steerable filter selects the horizontal direction. The amplitude signal is primarily confined to the region of the ball, while the phase signal disperses throughout the entire image space.

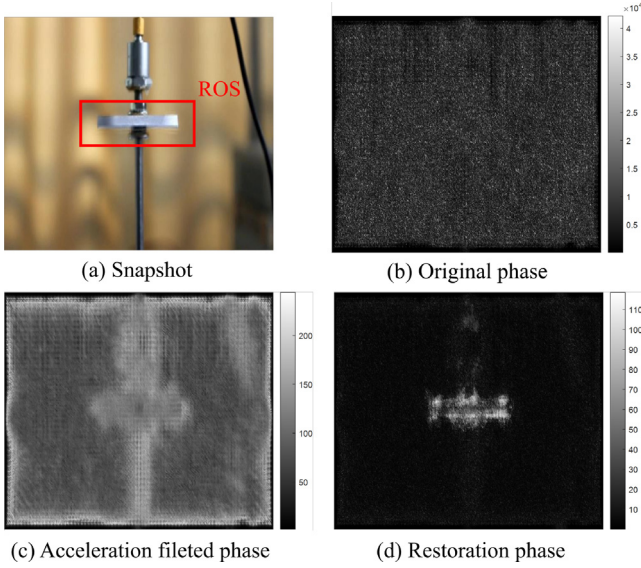


Fig. 4. The superimposed phase matrices at different steps of the proposed method. The superimposed phase matrix is obtained by summing the absolute phase differences between each frame and the first frame, showcasing the distribution of phase signals at each position in the video. (a) Snapshot of a vibrating exciter video. (b) Original phase matrix (after phase unwrapping). (c) Phase matrix after acceleration filtering. (d) Phase matrix after sparse restoration.

where $\phi_{\text{Mag}}^l(\mathbf{x}, t)$ is the magnified phase signal, and α is the magnification factor. Then, pyramid reconstruction is performed to generate the motion magnified video. Since our method only magnifies the subtle motion signals without amplifying other motion components, it enables the enhancement of the target's subtle motion under the interference of handheld camera movement.

Fig. 5 illustrates in detail the signal transformations at each step in the video space during the execution of the proposed method. The Video Decomposition step extracts amplitude and phase signals at different scales and orientations using the CSP. The Motion Separation step separates target subtle motion signals from camera motion signals through acceleration filtering and sparse reconstruction. The Video Reconstruction process amplifies the subtle motion signals and outputs the motion-magnified video.

4. Experiment results

To validate the effectiveness of the proposed method in magnifying subtle motions in videos recorded by handheld cameras, experiments

were conducted using real videos and synthetic videos. Qualitative and quantitative metrics were used to evaluate the performance of the proposed method and the comparison methods. The source videos and magnification results involved in the experiment can be found at <https://github.com/singleThreadD/PVMHC>.

4.1. Overall performance tests

The proposed method was compared with state-of-the-art approaches in the field of video magnification, including the foundational algorithm Phase-based (Wadhwa et al., 2013), acceleration filtering-related algorithms such as Acceleration (Zhang et al., 2017), Jerk-aware (Takeda et al., 2018), and BVMF (Takeda et al., 2022), as well as magnification with video stabilization as preprocessing (VS-Phase-based) (Alam et al., 2017). Qualitative and quantitative experiments were conducted. The qualitative experiments used real videos captured with handheld cameras as the material, where the ground truth for magnification was unknown. The quantitative experiments used artificially synthesized videos of a ball, where the magnification ground truth could be provided by controlling the vibration amplitude of the ball.

4.1.1. Qualitative analysis

Qualitative magnification experiments were conducted using a handheld Canon camera and a smartphone. The experimental subjects and camera parameters are shown in Table 1. Figs. 6, 7, and 8 show the magnification results of the exciter video, toy video, and cantilever beam video with a magnification factor of 20 under various comparison methods. The Phase-based (Wadhwa et al., 2013) method causes over-magnification of the target region, which contains both subtle motion and camera motion, and also magnifies the background camera motion. The Acceleration (Zhang et al., 2017) method produces the worst magnification results, with the subtle motion of the target appearing chaotic and the background camera motion significantly amplified. This is due to the erroneous magnification of a large amount of high-amplitude non-linear signals from the background camera motion. The Jerk-aware (Takeda et al., 2018) and BVMF (Takeda et al., 2022) methods show slight improvement in magnification results. The contours of the subtle motion in the target are clearer in the time slice images, and the background camera motion is not significantly amplified, as both methods suppress the high-amplitude non-linear signals. The VS-Phase-based method (Alam et al., 2017) suppresses subtle motion signals during the motion correction process, causing the generated magnified video to fail to accurately reflect the motion patterns of the source video. Although the SwinTransformer-based method (Lado-Roigé and Pérez, 2023) employs a neural network to learn the shape features after magnification, its network architecture is designed for videos captured by a stationary camera. As a result, it is not suitable for

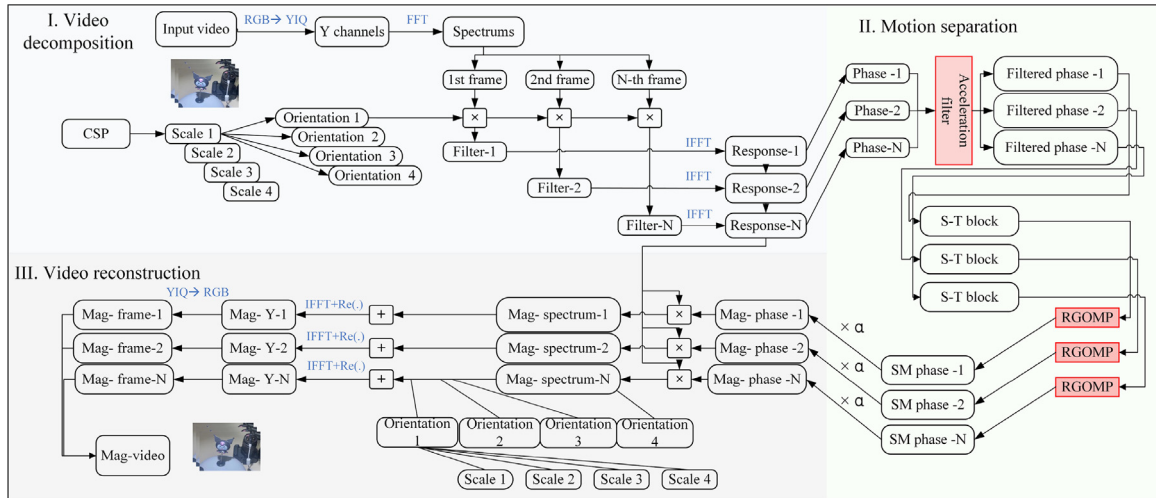


Fig. 5. Signal flow of the proposed method. FFT represents the Fast Fourier Transform, IFFT represents the Inverse Fast Fourier Transform, and $\text{Re}(\cdot)$ denotes taking the real part of a complex signal.

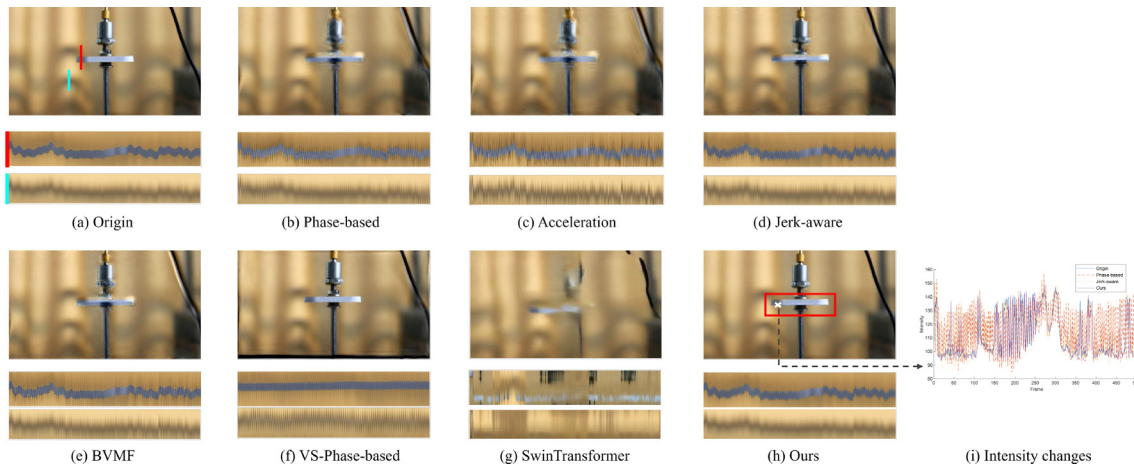


Fig. 6. The exciter video is magnified by various methods with the magnification factor of 20. (a) Original exciter video. Top: The frame selected from the original video. Bottom: The pixel-time slices at the red and cyan regions of the video. The red line segments are taken from the target exciter region, while the cyan line segments are taken from the background region. (b) Phase-based method (Wadhwa et al., 2013). (c) Acceleration method (Zhang et al., 2017). (d) Jerk-aware method (Takeda et al., 2018). (e) BVMF method (Takeda et al., 2022). (f) VS-Phase-based method (Alam et al., 2017). (g) SwinTransformer (Lado-Roigé and Pérez, 2023). (h) Our method. The red rectangle in (h) represents the ROS. (i) Intensity changes of the selected point. Since the videos used in the experiments are color images, intensity values were extracted after converting them to grayscale. For clarity of presentation, (i) only compares our method with the classic Phase-based method (Wadhwa et al., 2013) and the Jerk-aware method (Takeda et al., 2018), the latter of which demonstrated better performance. (b-e) all incorrectly magnified the background motion while amplifying the exciter movements, and the stabilization process in (f) caused the original motion pattern to be distorted. In the magnification results based on SwinTransformer in (g), the morphology of the target object exhibits distortion. In contrast, our approach successfully magnifies the subtle exciter movements without amplifying the background motion.

motion magnification under camera instability, leading to distortions in both the contours and shapes of the target objects in the magnified results. In addition, the network architecture does not incorporate prior knowledge of the target motion frequency band, making it difficult to accurately extract the subtle motion signals of interest. In contrast, our method accurately magnifies the regular subtle motions of the target while avoiding erroneous amplification of stationary background objects caused by camera motion. This advantage arises from the separation of subtle motion signals in the phase domain and the restriction of magnification to the spatial location of the target.

As shown in the intensity variations of the fixed points in Figs. 6(h), 7(h) and 8(h), our method amplifies only the small oscillations superimposed on the trend component without amplifying the trend component itself. It is important to note that the trend component is caused by camera motion. In contrast, the comparison methods indiscriminately

Table 1

Experimental subjects and camera parameters.

Object	Device	FPS	H*W	T	TF (Hz)	α
Exciter	Canon EOS	50	640*360	502	10.2	20
Toy	iPhone 11	30	558*362	279	7	20
Cantilever beam	iPhone 11	30	960*544	337	4.5	20

Note: H: height of the video frame. W: width of the video frame. T: total frames of the video. TF: target frequency. α : magnification factor.

amplify the components of the selected frequency band within the trend component, thereby incorrectly magnifying the camera motion.

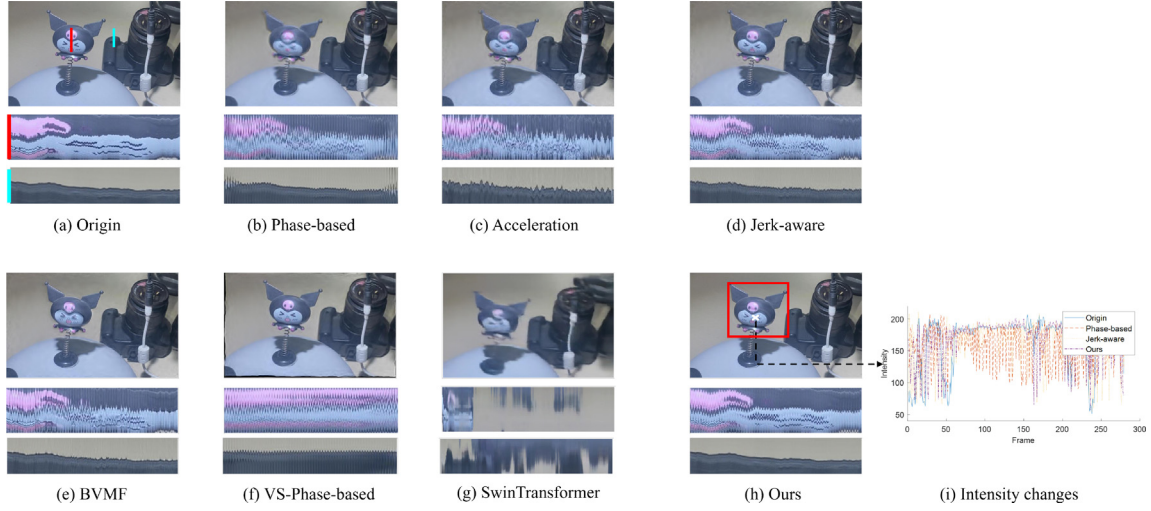


Fig. 7. The toy video is magnified by various methods with the magnification factor of 20. (a) Original toy video. Top: The frame selected from the original video. Bottom: The pixel-time slices at the red and cyan regions of the video. The red line segments are taken from the target toy region, while the cyan line segments are taken from the background region. (b) Phase-based method (Wadhwa et al., 2013). (c) Acceleration method (Zhang et al., 2017). (d) Jerk-aware method (Takeda et al., 2018). (e) BVMF method (Takeda et al., 2022). (f) VS-Phase-based method (Alam et al., 2017). (g) SwinTransformer (Lado-Roigé and Pérez, 2023). (h) Our method. The red rectangle in (h) represents the ROS. (i) Intensity changes of the selected point. (b-e) while amplifying the target toy movement, all incorrectly magnified the “motion” of stationary objects in the background. The stabilization process in (f) caused the original motion pattern to be distorted. In the magnification results based on SwinTransformer in (g), the morphology of the target object exhibits distortion. In contrast, our method successfully magnifies the subtle toy movements without amplifying the background motion.

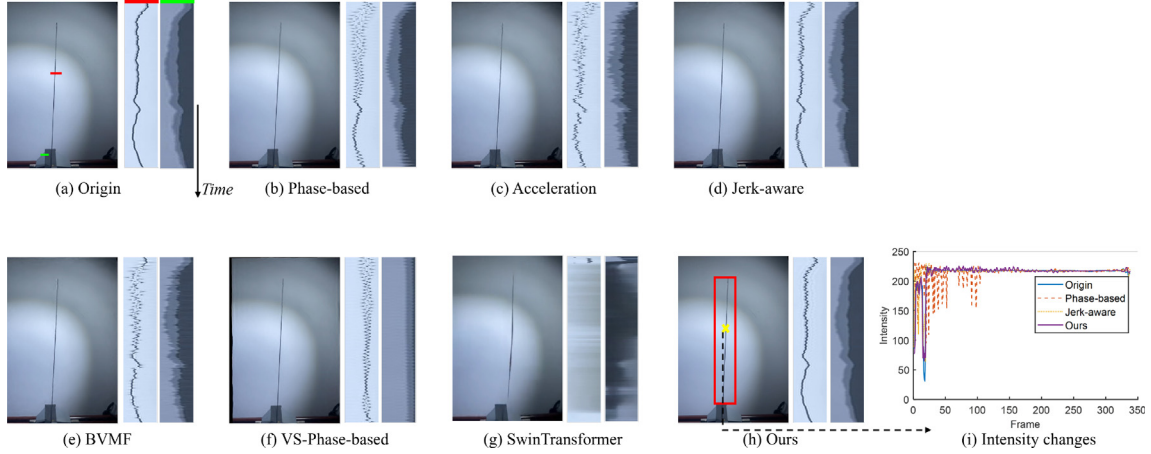


Fig. 8. The vibrating cantilever beam video is magnified by various methods with the magnification factor of 20. (a) Original cantilever beam video. Top: The frame selected from the original video. Bottom: The pixel-time slices at the red and green regions of the video. The red line segments are taken from the target cantilever beam region, while the green line segments are taken from the background region. (b) Phase-based method (Wadhwa et al., 2013). (c) Acceleration method (Zhang et al., 2017). (d) Jerk-aware method (Takeda et al., 2018). (e) BVMF method (Takeda et al., 2022). (f) VS-Phase-based method (Alam et al., 2017). (g) SwinTransformer (Lado-Roigé and Pérez, 2023). (h) Our method. The red rectangle in (h) represents the ROS. (i) Intensity changes of the selected point. (b-e) while amplifying the vibration of the cantilever beam all incorrectly magnified the “motion” of stationary objects in the background. The stabilization process in (f) caused the original motion pattern to be distorted. In the magnification results based on SwinTransformer in (g), the morphology of the target object exhibits distortion. In contrast, our method successfully magnifies the subtle vibration without amplifying the background motion.

4.1.2. Quantitative analysis

To quantitatively verify the superiority of the proposed method, a synthetic simulation of a vibrating ball video is conducted to carry out controlled motion magnification experiments. The production of the ball video mainly involves two steps: (1) Generating sub-pixel level micro-vibrations of the ball. This is achieved by first generating a high-resolution video and then downsampling it. After downsampling, Gaussian filtering is applied to smooth the video. (2) Generating a camera shake effect. Random X and Y translation displacements between every two frames are generated and superimposed onto the vibrating ball video. In this experiment, the vibration frequency of the ball is set to 10 Hz, the amplitude is set to 0.25 pixels, and the maximum pixel displacement between two frames caused by camera shake is set

to 3 pixels. The vibrating ball is magnified by various methods with a magnification factor of 10.

Fig. 9 shows the pixel-time slices of the original video, the ground truth magnified video, and the magnified videos generated by various comparison methods. With a magnification factor of 10, the ground truth can be generated simply by adjusting the vibration amplitude of the ball to 2.5 pixels. The results show that the Phase-based (Wadhwa et al., 2013), Acceleration (Zhang et al., 2017), Jerk-aware (Takeda et al., 2018), and BVMF (Takeda et al., 2022) methods all significantly amplified the false motion of the textures in the background, and also introduced artifacts and glitches in the motion magnification of the ball itself. The VS-Phase-based (Alam et al., 2017) method altered the original motion pattern. Our approach accurately magnifies the subtle

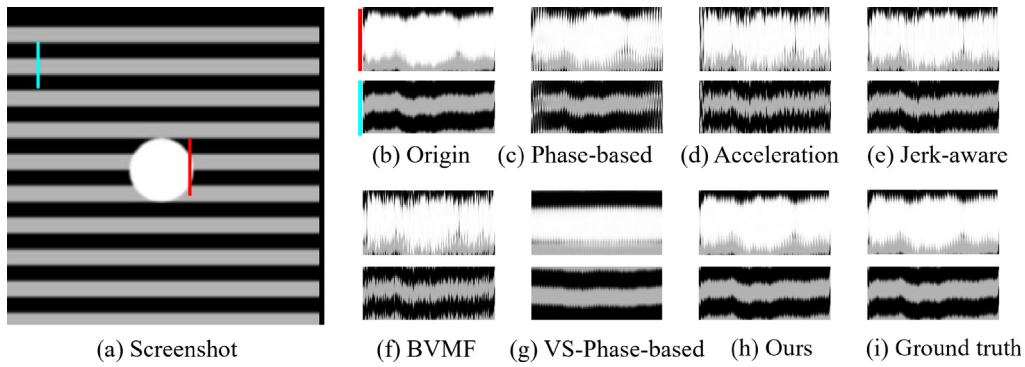


Fig. 9. The simulated ball video is magnified by various methods with the magnification factor of 10. (a) Screenshot of the original ball video. (b) Pixel-time slices of the original ball video. The red line segments are taken from the target ball region, while the cyan line segments are taken from the background region. (c) Phase-based method (Wadhwa et al., 2013). (d) Acceleration method (Zhang et al., 2017). (e) Jerk-aware method (Takeda et al., 2018). (f) BVMF method (Takeda et al., 2022). (g) VS-Phase-based method (Alam et al., 2017). (h) Our method. (i) Ground truth: pixel-time slices of the true magnified video. (c-f) all incorrectly magnified the background motion while amplifying the ball movements. The stabilization process in (g) caused the original motion pattern to be distorted. In contrast, our method demonstrates a high level of consistency with the ground truth because it successfully magnifies the subtle ball movements without amplifying the background motion.

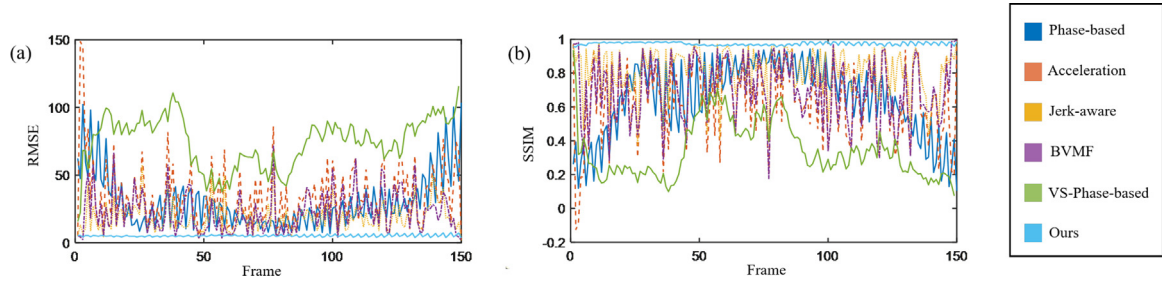


Fig. 10. Quantitative evaluation metrics for the magnified videos of each method compared with the ground truth magnified videos in the simulated ball experiment. (a) Root Mean Square Error (RMSE) of each frame. (b) Structural Similarity Index (SSIM) of each frame.

movements of the target ball, closely resembling the pixel-time slices of the ground truth magnification.

Fig. 10 shows the root mean square error (RMSE) and structural similarity index (SSIM) between the magnified videos of each method and the ground truth video for each frame. Lower RMSE and higher SSIM indicate better magnification quality. It can be seen that our approach consistently maintains stable magnification quality throughout the entire duration of the video, outperforming the comparison methods.

4.2. Ablation tests

To verify the effectiveness of RGOMP-based phase reconstruction for micro-motion magnification, a set of ablation experiments is conducted using a simulated vibrating ball video. The comparison groups are designed as four different combinations of the key steps in the proposed method: (1) Jerk: Only the full-field phase is processed using Jerk-aware acceleration filtering (Takeda et al., 2018), which is equivalent to the Jerk-aware method in the previous comparison experiments. (2) Jerk+OMP: The filtered full-field phase is further reconstructed using the classical OMP method. (3) Jerk+WOMP: The reconstruction of the full-field phase is performed using a weighted OMP strategy. Specifically, the phase values within the manually selected region are assigned a weight of 1, while those outside the region are assigned a weight of (η). In this experiment, η is set to 0.5. (4) Jerk+RGOMP: The reconstruction of the full-field phase is carried out using a weighted RGOMP strategy (i.e., the complete proposed method), where adaptive selection of phase points is further performed under the guidance of the manually selected region. The parameter η is also set to 0.5. The experimental video is a simulated vibrating ball. The vibration frequency,

amplitude, and magnification factor are set to 10 Hz, 0.25 pixels, and 10 $\times$, respectively. The maximum pixel displacement between two consecutive frames caused by camera shake is set to 5 pixels.

Fig. 11 presents the pixel-time slices of the original video, the ground-truth magnified video, and the magnified videos generated by each comparison group. For the Jerk group, jagged and discontinuous variations appear in the target region, while significant camera motion amplification is observed in the background. By introducing OMP for phase selection and reconstruction (Jerk+OMP), both the jagged artifacts and false amplification are alleviated. With the further introduction of weighted reconstruction (Jerk+WOMP), the false amplification in the background is almost eliminated. Finally, using RGOMP-based phase reconstruction (Jerk+RGOMP), i.e., the complete proposed method, the jagged artifacts in the target region nearly disappear, and the overall motion magnification performance reaches the best level. The quantitative metrics shown in Table 2, including the RMSE and SSIM with respect to the ground-truth magnified video, also demonstrate that the magnified video generated by Jerk+RGOMP is the most consistent with the ground truth.

The experimental results indicate that OMP has a certain capability in suppressing full-field phase variations; however, since it does not incorporate region information of the vibrating target, a considerable amount of camera motion components remains. WOMP introduces a regional weighting mechanism, but the manually selected region still contains a large amount of camera-induced phase variations, limiting its ability to suppress interference. In contrast, RGOMP, guided by the manually selected region, performs a more refined selection of phase points within the region, preserving the phase information of the target micro-motion to the greatest extent while avoiding redundant background camera motion. As a result, it achieves the best motion magnification performance.

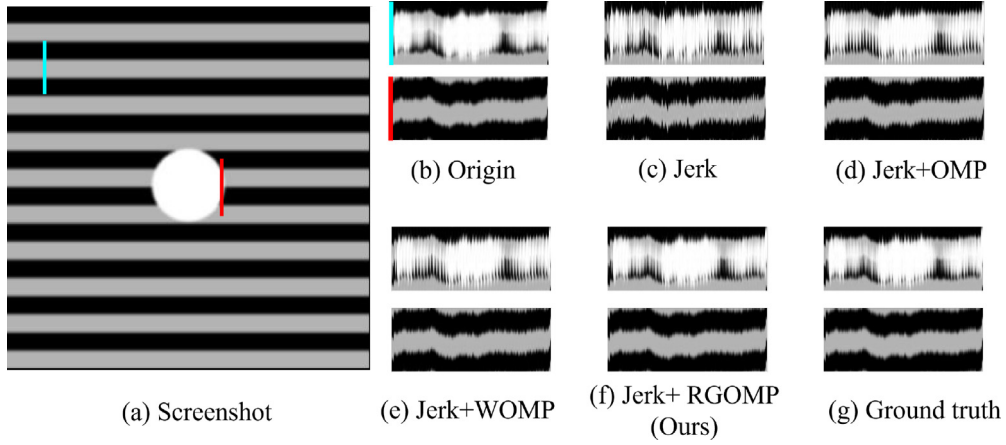


Fig. 11. The simulated ball video is magnified by each ablation group with the magnification factor of 10. (a) Screenshot of the original ball video. (b) Pixel-time slices of the original ball video. The red line segments are taken from the target ball region, while the cyan line segments are taken from the background region. (c) Jerk. (d) Jerk+OMP. (e) Jerk+WOMP. (f) Jerk+RGOMP (Ours). (g) Ground truth: pixel-time slices of the true magnified video.

Table 2
Ablation tests.

Method	Jerk	Jerk+OMP	Jerk+WOMP	Jerk+RGOMP (Ours)
RMSE	18.8037	14.0711	9.7687	5.3891
SSIM	0.8138	0.8848	0.9270	0.9718

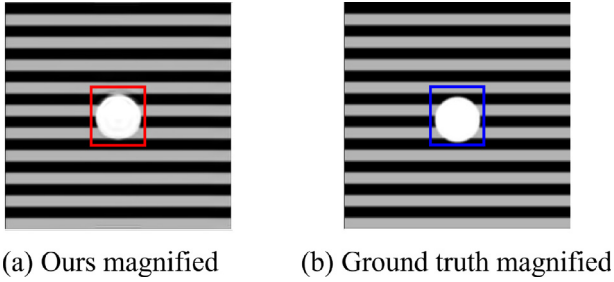


Fig. 12. Division of target subtle motion area and background area. (a) Magnified video produced by our method. The red rectangle indicates the target subtle motion area, and the area outside it represents the background. (b) Ground truth magnified video generated through simulation. The blue rectangle indicates the target subtle motion area, and the area outside it represents the background. The magnification effect is finely evaluated by comparing the RMSE of the corresponding areas in (a) and (b).

4.3. Key parameter tests

Our method involves two key parameters in the process of magnifying subtle motions: the size of the ROS and the background retention coefficient (η) defined in Eq. (19). The former determines the area within which subtle motion phases are selected, while the latter relates to the extent of subtle motion retention in the background (or, conversely, the degree of suppression of camera motion in the background). We conducted tests on a simulated ball dataset containing five levels of camera instability. The ball's vibration frequency was set to 10 Hz, with an amplitude of 0.25 pixels. The degree of camera instability was controlled by setting the maximum displacement of simulated camera motion between frames to 1, 3, 5, 7, and 9 pixels, respectively. To precisely evaluate the impact of key parameters in the proposed method, we calculated the RMSE between the subtle motion region of the output video (i.e., the ball region) and the background region against the ground truth magnified video. The background region and target region are manually specified by us, as shown in Fig. 12. To

verify the impact of the ROS area, we set 10 different sizes of ROS and recorded the RMSE of the target region and background under each ROS, as shown in Fig. 13. To evaluate the influence of the background retention coefficient, we varied it from 0 to 1 with an interval of 0.1 and recorded the evaluation metrics during the process. The results are shown in Fig. 14.

The results show that as the ROS area increases, the RMSE between the target region and the ground truth magnified video gradually decreases (except for a slight increase at points 5, 6, and 7), indicating that a larger ROS area improves the quality of subtle motion magnification. This is because a larger ROS area can recover more subtle motion phase signals. In contrast, the RMSE between the background region and the ground truth magnified video gradually increases. This is due to the fact that as the ROS area expands, the camera motion signals in the background region are also magnified more. This demonstrates the trade-off between preserving subtle motion signals and suppressing camera motion. To ensure the overall quality of the output magnified video, it is recommended to select a moderately sized ROS in practice, generally slightly larger than the target region.

As the background retention coefficient changes, the trade-off between the quality of subtle motion magnification and the suppression of camera motion interference becomes evident. When the background retention coefficient increases, the subtle motion magnification quality improves due to reduced loss of subtle motion phase signals dispersed in the background. On the other hand, since the camera motion phases in the background are retained to a greater extent, the interference from camera motion is also magnified more significantly, resulting in more artifacts appearing in the background. It should be noted that, to balance this trade-off, the background retention coefficient (η) of the proposed method was set to 0.5 in both the previous overall performance tests and the subsequent influencing factor tests.

4.4. Influencing factors tests

To thoroughly analyze the performance boundaries and applicable scenarios of the proposed method, we conducted tests on the impact of magnification factor and the degree of camera instability on video magnification quality. The comparison methods mentioned in the overall performance tests were also included in these tests.

4.4.1. Impacts of magnification factors

Fig. 15 shows the magnified videos of the cantilever beam vibration at different magnification factors. Since the Acceleration (Zhang et al., 2017) and VS-Phase-based (Alam et al., 2017) methods performed poorly in the overall performance tests, they are not included in the

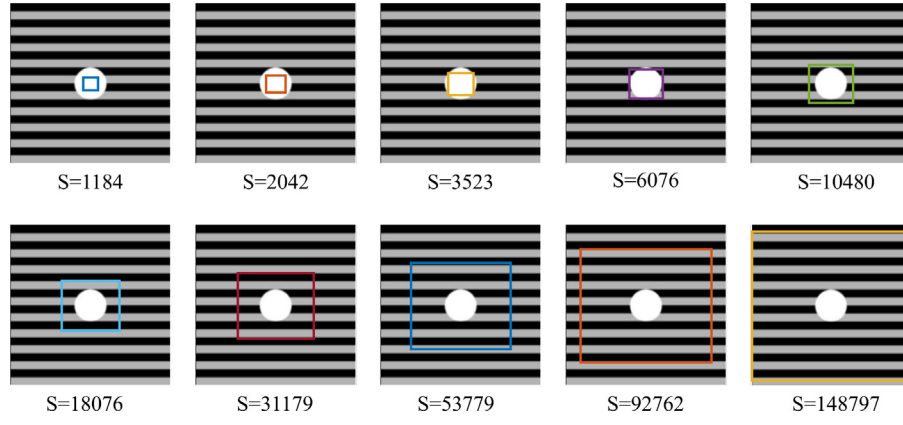


Fig. 13. Different ROS of the simulated ball video. Rectangles of different colors represent ROSs of different sizes, and the area of the ROS is marked as S at the bottom of each sub-image.

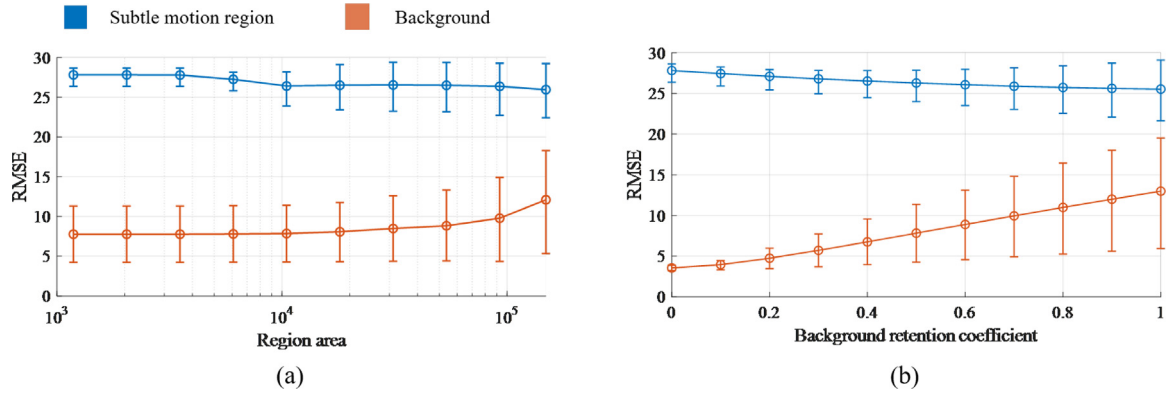


Fig. 14. Magnification effect metrics under different ROS areas and background retention coefficients.

comparison here. In the cantilever beam region, as the magnification factor increases, the amplified motions generated by each method exhibit larger amplitudes. For Phase-based (Wadhwa et al., 2013) and BVMF (Takeda et al., 2022), significant artifacts appear in the target region when the magnification factor reaches 20 or 30. This is because both methods amplify the phases of subtle motion and camera motion simultaneously, exceeding the displacement limit denoted in Section 2 (Assume that the magnification factor is fixed). Jerk-aware (Takeda et al., 2018) and our method achieve more reliable magnification results in the target region. In the background region, Phase-based (Wadhwa et al., 2013), BVMF (Takeda et al., 2022), and Jerk-aware (Takeda et al., 2018) amplify the pseudo-motion caused by the camera, and the amplitude of the pseudo-motion in the output video increases as the magnification factor grows. In contrast, our method does not mistakenly amplify the pseudo-motion caused by the camera under any of the three magnification factors.

To fully evaluate the impact of magnification factors, quantitative tests are conducted using a simulated ball video dataset. The ball vibration frequency is set to 10 Hz, with a vibration amplitude of 0.25 pixels, and the degree of camera instability is set to 5 levels. The camera instability is controlled by the maximum pixel shift between consecutive frames, with these 5 levels corresponding to 1, 3, 5, 7, and 9 pixels, respectively. The ball video is magnified by 5 to 60 times (in increments of 5) using the proposed method and comparison methods, and the RMSE and SSIM with the ground truth magnified video are calculated. The ground truth is provided by controlling the vibration amplitude of the ball.

Fig. 16(a)(b) shows the comparison results of different methods under the influence of magnification factors. As the magnification factor varies from 5 to 60, the performance of the Phase-based (Wadhwa

et al., 2013), Acceleration (Zhang et al., 2017), and BVMF (Takeda et al., 2022) methods declines rapidly and eventually stabilizes. This is because these methods fail to effectively separate the target subtle motion from the overall camera offset signals in the phase space. The performance of the Jerk-aware method slightly improves compared to the previous three methods, but its RMSE still quickly exceeds 20. The VS-Phase-based (Alam et al., 2017) method introduces distortion to the subtle motion in the video due to its stabilization process, resulting in consistently high RMSE values (or low SSIM) compared to the ground truth magnified video, with little variation across magnification factors. Our method outperforms the Jerk-aware (Takeda et al., 2018) method and demonstrates better performance as the magnification factor increases. This is because our method reconstructs the phase of subtle motion in both the temporal and spatial domains. As the magnification factor varies from 5 to 60, our method achieves RMSE and SSIM values of 18.33 and 0.8594, respectively, representing improvements of 61.6% and 67.6% compared to the Jerk-aware (Takeda et al., 2018) method.

It should be noted that, under strictly static camera conditions, classical phase-based motion magnification methods can theoretically support much larger magnification factors. However, in practical scenarios with camera instability, the admissible magnification range is fundamentally constrained by the total displacement encoded in the phase. As discussed in Section 2, the magnification factor is bounded by the phase displacement limit. When camera motion is present, the observed phase variation includes both target vibration and camera-induced motion, which rapidly drives the amplified displacement beyond this limit at large magnification factors. This explains the severe artifacts observed in existing methods and further highlights the advantage of the proposed approach in maintaining stable magnification performance under increasing magnification factors and camera instability.

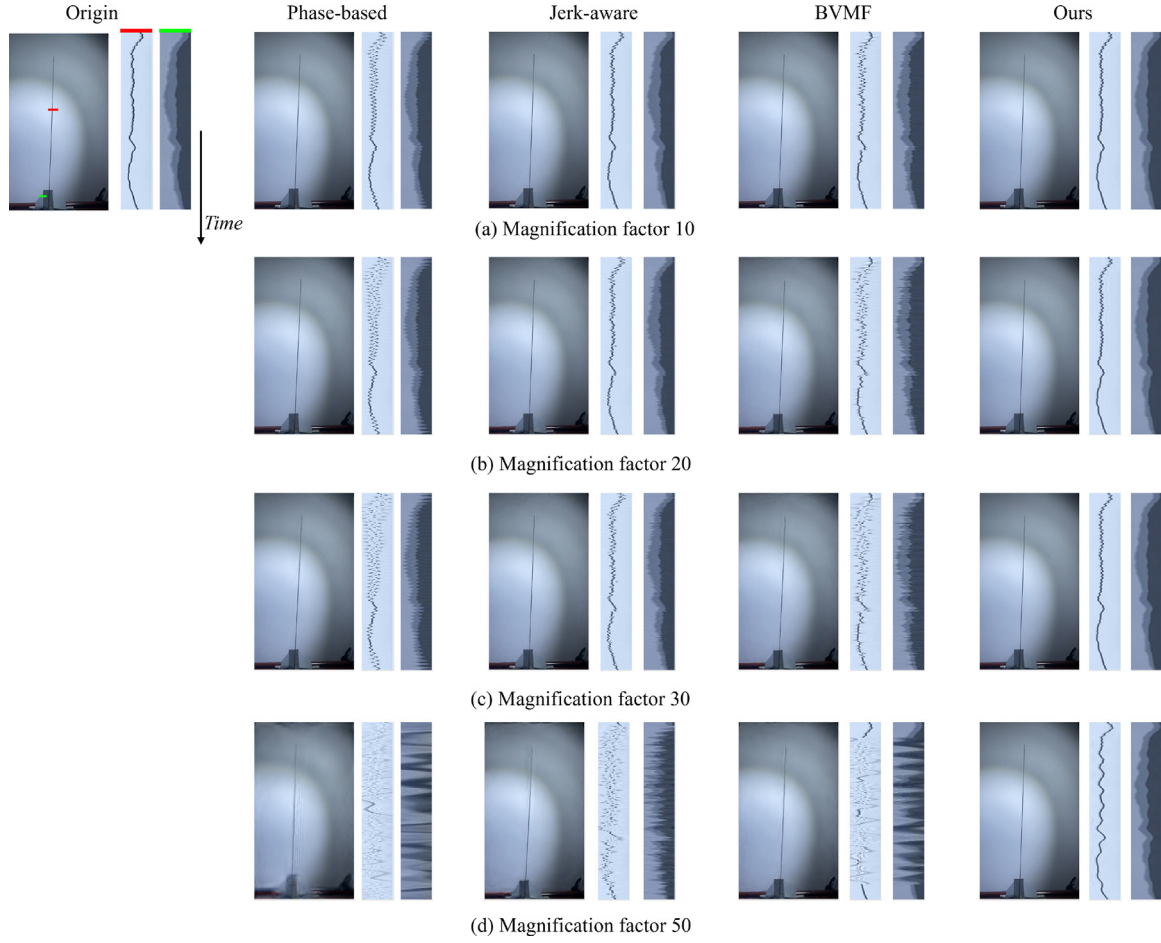


Fig. 15. The cantilever beam video is magnified by the Phase-based (Wadhwa et al., 2013), Jerk-aware (Takeda et al., 2018), BVMF (Takeda et al., 2022) and our approach with the magnification factors of 10, 20, 30 and 50. (a) Magnification results with the factor 10. (b) Magnification results with the factor 20. (c) Magnification results with the factor 30. (d) Magnification results with the factor 50. The pixel-time slices still represent the magnification results of the target subtle motion region (red) and the background region (green).

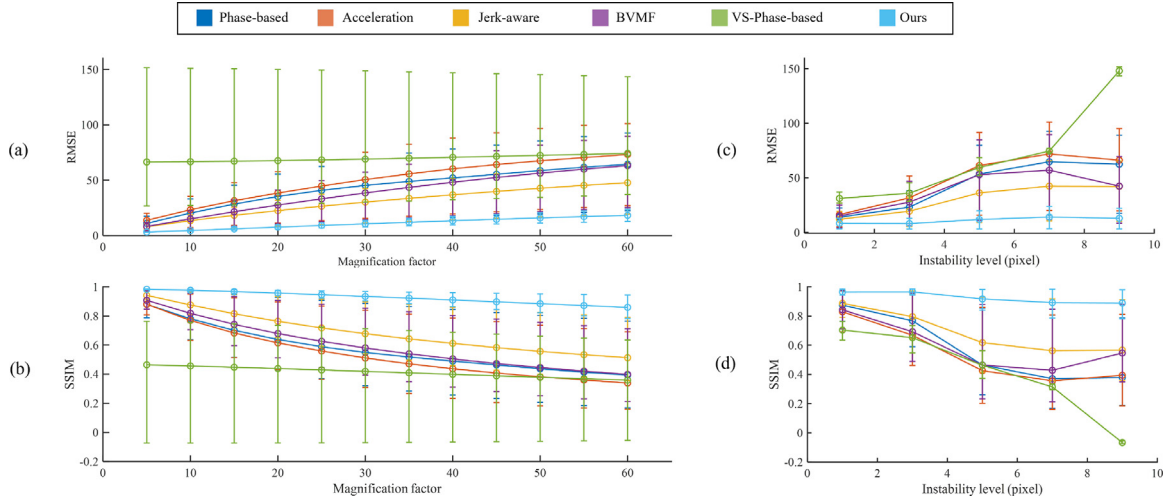


Fig. 16. The quantitative test results showing the effects of the magnification factors and the instability levels. (a) RMSE results for magnification factors from 5 to 30. (b) SSIM results for magnification factors from 5 to 30. (c) RMSE results for various camera instability levels. (d) SSIM results for various camera instability levels.

4.4.2. Impacts of camera instability

Fig. 17 shows the magnified videos of the subtle motions of the exciter under different levels of camera instability. The degree of camera instability is controlled by hand tremors during recording.

As the degree of camera instability increases, Phase-based (Wadhwa et al., 2013), Jerk-aware (Takeda et al., 2018), and BVMF (Takeda et al., 2022) produce more significant artifacts in the target region. This is because, with higher instability, the background region of the

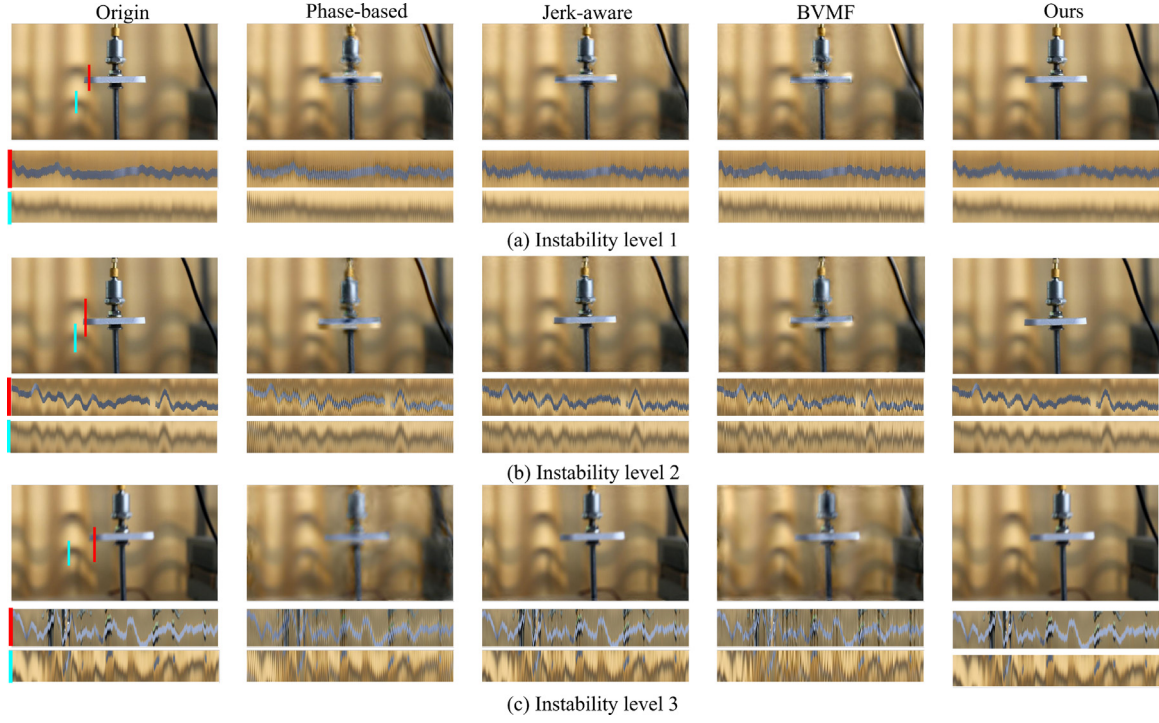


Fig. 17. The exciter video is magnified by the Phase-based (Wadhwa et al., 2013), Jerk-aware (Takeda et al., 2018), BVMF (Takeda et al., 2022) and our approach with different instability levels. The degree of camera instability is controlled by hand tremors during recording. (a) Magnification results with instability level 1. (b) Magnification results with instability level 2. (c) Magnification results with instability level 3. The pixel-time slices still represent the magnification results of the target subtle motion region (red) and the background region (cyan).

video generates richer pseudo-motion signals, leading to more severe over-magnification effects in the background for Phase-based (Wadhwa et al., 2013), Jerk-aware (Takeda et al., 2018), and BVMF (Takeda et al., 2022). Our method can still achieve subtle motion magnification under the first two levels of instability but shows less effective magnification of the exciter's subtle vibrations at level 3. The reason is that when the hand trembles violently (with higher instability), the subtle vibration signals mix with the camera motion, appearing as intense, large-amplitude signals. These are subsequently removed by the Jerk filter, resulting in the phase reconstruction process recovering only a small portion of the signal. Therefore, an increase in camera instability gradually renders our method less effective.

To fully evaluate the impact of camera instability, quantitative tests were conducted using the simulated ball video dataset from the magnification factor experiment. In this dataset, five levels of instability were created by setting the maximum pixel shift to 1, 3, 5, 7, and 9. For each instability level, the magnification factor varied from 5 to 60 in steps of 5, resulting in a total of 12 video sequences. Fig. 16(c)(d) shows the magnification performance of various methods under different levels of instability. It can be observed that as the maximum pixel shift between frames increases, i.e., as the degree of camera instability increases, the magnification performance of Phase-based (Wadhwa et al., 2013), Acceleration (Zhang et al., 2017), Jerk-aware (Takeda et al., 2018), BVMF (Takeda et al., 2022), and our method shows an overall declining trend. However, when the instability level reaches 9 pixels, there is a slight improvement. The reason lies in the periodic nature of phase. When the total motion signal in the video becomes sufficiently large, the artifacts caused by magnification do not necessarily become more severe, and the video quality does not necessarily continue to degrade but instead stabilizes at a certain level. For the VS-Phase-based method (Alam et al., 2017), as the degree of instability increases, the stabilization process applied by the algorithm causes the motion patterns in the output video to deviate further from the ground truth.

Overall, both the magnification factor and the degree of camera instability affect the quality of the magnified video. As the magnification

factor increases and the degree of camera instability rises, the magnification quality of our method gradually decreases. It should be noted that due to the interference from camera motion, the magnification factor in handheld camera scenarios is difficult to reach the levels achieved by stable cameras (in studies on phase-based magnification methods, for example, the magnification factor typically reaches 100–200 times). This is an area for future research to address.

4.5. Computational complexity analysis

In this section, we analyze the computational complexity of the proposed method with respect to the spatial resolution and temporal length of the input video. Let the spatial resolution be $H \times W$, the number of frames be T , and $N = H \times W$ denote the total number of pixels. Let L and O represent the number of pyramid levels and orientations, K the sparsity level in RGOMP, and M the dictionary size. The pipeline consists of three stages: (1) multi-scale phase decomposition, (2) temporal phase processing, and (3) sparse phase reconstruction.

In the first stage, applying 2D FFT to each frame results in a complexity of $\mathcal{O}(T \cdot N \log N)$, followed by pyramid decomposition with cost $\mathcal{O}(L \cdot O \cdot T \cdot N)$. The second stage, including phase extraction, unwrapping, and temporal filtering, also incurs $\mathcal{O}(L \cdot O \cdot T \cdot N)$ complexity. The third stage dominates the computation: treating each pixel as a temporal signal of length T , sparse reconstruction via OMP/RGOMP leads to $\mathcal{O}(N \cdot K \cdot M \cdot T)$ complexity. Overall, the total complexity can be summarized as

$$\mathcal{O}(T \cdot N \log N + L \cdot O \cdot T \cdot N + N \cdot K \cdot M \cdot T), \quad (23)$$

where the sparse reconstruction term is dominant, especially for large N , T , or M . The space complexity is approximately $\mathcal{O}(N \cdot T + N \cdot M)$. In practice, the method scales near-linearly with spatial resolution and temporal length, while RGOMP forms the main computational bottleneck.

To validate this analysis, we conduct runtime experiments across different methods, resolutions, and frame counts. As shown in

Table 3
Runtime comparison of different methods.

	Method					
	Phase-based	Acceleration	Jerk-aware	BVMF	VS-Phase-based	Ours
Runtime (s)	78.27	69.92	134.72	69.88	93.39	334.82
Setup	AMD Ryzen 7 5800H with 32 GB RAM, NVIDIA RTX 3060 GPU (6 GB), MATLAB R2021a					
Video size	640px × 360px × 500					

Table 4
Runtime under different resolutions and frame counts (unit: s).

Resolution	Frame counts				
	100 frames	200 frames	300 frames	400 frames	500 frames
128px × 72px	7.78	12.49	17.00	20.72	26.72
256px × 144px	18.71	29.37	39.92	50.23	62.18
384px × 216px	36.83	55.59	75.55	93.80	113.88
512px × 288px	62.76	96.61	133.07	165.76	210.61
640px × 360px	97.46	148.39	205.04	269.33	334.82

Table 5
Runtime breakdown of different optimization strategies (unit: s).

Module	Original proposed method	DC	DC+parfor	DC+parfor+ROI
preprocess	6.97	7.05	7.00	7.03
FFT_CSF	0.97	1.03	1.00	1.02
Accel. filtering	91.87	94.13	94.59	93.26
RGOMP	201.61	98.77	66.11	50.28
rendering	6.86	6.06	5.19	5.11
Total	308.28	207.04	173.89	156.70

Table 3, the proposed method requires more runtime than existing approaches due to the iterative sparse reconstruction, whereas prior methods mainly rely on linear temporal filtering. **Table 4** further shows that the runtime increases consistently with both the frame count T and the number of pixels N , which generally agrees with the theoretical complexity analysis. However, the growth rate is slightly sub-linear in practice. This behavior can be attributed to several factors, including fixed computational overhead (e.g., preprocessing and memory operations), improved parallel efficiency for larger workloads, and optimized matrix operations in MATLAB that reduce per-element computational cost at larger scales. As a result, the empirical runtime exhibits a slower-than-linear increase while still maintaining overall scalability with respect to T and N .

To alleviate the computational burden, we explore several practical acceleration strategies, with results summarized in **Table 5**. Specifically, we consider: (1) dictionary compression (DC) via low-rank approximation to reduce M ; (2) parallel implementation using *parfor* to exploit pixel-wise independence; and (3) ROI-guided processing to avoid redundant computation in background regions. The RGOMP stage dominates runtime in the original implementation. With DC, its runtime decreases from 201.61 s to 98.77 s, confirming the impact of dictionary size. Incorporating parallelization further reduces it to 66.11 s, and ROI-based processing lowers it to 50.28 s by suppressing unnecessary computations. It should be noted that the total runtime of the original proposed method reported in **Table 5** differs from that in **Tables 3** and **4**. This discrepancy arises because the latter results are obtained under batch-processing conditions, where sequential execution introduces additional overhead (e.g., resource contention and frequency scaling), whereas **Table 5** reports isolated profiling results. Nevertheless, the relative performance trends remain consistent.

Overall, these strategies reduce total runtime from 308.28 s to 156.70 s, achieving nearly a 2× speedup without modifying the core framework. This demonstrates that the method, while computationally intensive, can be significantly accelerated and is suitable for offline high-quality analysis. However, ROI-based acceleration introduces a limitation. Restricting reconstruction to foreground regions may suppress subtle motion-related phase variations diffused into the background. As illustrated in **Fig. 3**, such background phase components can still carry useful motion information due to spatial phase propagation. Therefore, completely excluding background regions is equivalent to setting the background retention coefficient (η) to zero (Section 4.3), potentially discarding weak but informative motion cues. In practice, using a moderate background retention ($\eta > 0$) provides a better balance between efficiency and reconstruction accuracy.

4.6. Limitations analysis

While the proposed method demonstrates strong performance in subtle motion magnification under handheld conditions, several limitations should be acknowledged.

(1) Dependence on manual ROI selection. The current framework relies on manually specified regions of interest (ROI) to guide motion extraction. Although the proposed RGOMP further refines motion representation within the selected region, the initial ROI still plays a critical role in determining the final reconstruction quality. Inaccurate or suboptimal ROI selection may include irrelevant motion components or exclude useful signals, thereby affecting the results. This limitation reduces the level of automation and may restrict applicability in large-scale or real-time scenarios. Future work could explore integrating automatic ROI detection, motion saliency estimation, or learning-based region proposal methods to alleviate this dependency.

(2) High computational cost. The method is computationally intensive, primarily due to the RGOMP-based sparse reconstruction, which operates on a per-pixel temporal basis across multiple scales and orientations. As analyzed in Section 4.5, this step dominates the overall runtime and limits scalability for high-resolution or long-duration videos. Although several acceleration strategies (e.g., dictionary compression, parallel processing, and ROI-guided computation) significantly reduce runtime, the method remains more suitable for offline processing rather than real-time applications.

(3) Trade-off between efficiency and motion fidelity. As discussed in Section 4.5, acceleration strategies such as ROI-based processing may introduce a trade-off between computational efficiency and reconstruction accuracy. In particular, aggressively excluding background regions may suppress weak but informative motion signals due to phase propagation effects. Although a moderate background retention strategy can mitigate this issue, it remains a design trade-off that depends on specific application scenarios.

Overall, these limitations highlight potential directions for future improvement, including automation, computational optimization, and further efficiency enhancement.

5. Conclusion

This paper proposes a method for magnifying subtle motions in videos captured by handheld cameras. To separate target subtle motions from camera motion interference, acceleration filtering is employed to initially suppress large-scale camera motion. Furthermore, leveraging the sparsity of subtle motions in the frequency domain and their local concentration in the spatial domain, a sparse coding framework is adopted, with the RGOMP algorithm designed to reconstruct regular subtle motion signals and eliminate residual camera interference. Experiments conducted on both real and synthetic video datasets demonstrate that the proposed method effectively magnifies subtle motions on handheld camera platforms and outperforms state-of-the-art methods in terms of magnification performance. Additionally, the key parameters of the proposed method, the influencing factors of magnification, and the computational complexity are systematically analyzed and discussed. The proposed method expands the application boundaries of motion magnification technology, making it possible to conveniently perform video motion magnification on mobile camera platforms such as smartphones. However, the method has certain limitations, such as reliance on manually selecting the target region and relatively high computational cost. Future work will focus on improving the level of automation and further optimizing computational efficiency.

CRediT authorship contribution statement

Zongdi Zang: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xuezhi Yang:** Funding acquisition, Conceptualization. **Gang Zhang:** Writing – review & editing, Software, Resources, Project administration. **Qin Xu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Xuezhi Yang reports financial support was provided by Science and Technology Department of Anhui Province. Qin Xu reports financial support was provided by Science and Technology Department of Gansu Province. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Major Science and Technology Project of Anhui Provincial Science and Technology Innovation Platform under Grant S202305a12020014, and was also supported by the Gansu Provincial Science and Technology Program under Grant 25JRRA225.

Data availability

The data used in this work is publicly available in <https://github.com/singleThreadD/PVMHC>.

References

- Al-Naji, A., Chahl, J., 2016. Remote respiratory monitoring system based on developing motion magnification technique. *Biomed. Signal Process. Control.* 29, 1–10.
- Alam, S., Singh, S.P.N., Abeyratne, U., 2017. Considerations of handheld respiratory rate estimation via a stabilized video magnification approach. In: 2017 39th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). pp. 4293–4296.
- Cai, E., Zhang, Y., Lu, X., Ji, X., Hou, J., Li, P., Guo, W., 2024. Vision-based automatic structural motion estimation in presence of 3D camera motion effects. *Autom. Constr.* 159, 105275.
- Cai, W., Zhao, J., Yi, R., Yu, M., Duan, F., Pan, Z., Liu, Y.-J., 2024. MFDAN: Multi-level flow-driven attention network for micro-expression recognition. *IEEE Trans. Circuits Syst. Video Technol.* 1, early access.
- Chen, L., Peng, C., Zhao, B., 2023. Novel multi-task learning for motion magnification. *IEEE Trans. Circuits Syst. Video Technol.* 33 (10), 5973–5985.
- Chen, J.G., Wadhwa, N., Cha, Y.-J., Durand, F., Freeman, W.T., Buyukozturk, O., 2015. Modal identification of simple structures with high-speed video using motion magnification. *J. Sound Vib.* 345, 58–71.
- Davis, A., Bouman, K.L., Chen, J.G., Rubinstein, M., Durand, F., Freeman, W.T., 2015. Visual vibrometry: Estimating material properties from small motion in video. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. pp. 5335–5343.
- Elgharib, M., Hsieh, M.H., Shechtman, A., Wetzstein, A.M., Matusik, W., 2015. Video magnification in presence of large motions. *ACM Trans. Graph.* 34 (6), 1–8.
- Feng, B.Y., Alzayer, H., Rubinstein, M., Freeman, W.T., Huang, J.-b., 2023. 3D motion magnification: Visualizing subtle motions from time-varying radiance fields. In: *Proc. IEEE/CVF Int. Conf. Computer Vision. ICCV*, pp. 9837–9846.
- Fioriti, V., Roselli, I., Cataldo, A., Forliti, S., Colucci, A., Baldini, M., Picca, A., 2022. Motion magnification applications for the protection of Italian cultural heritage assets. *Sensors* 22 (24), 9988.
- Fioriti, V., Roselli, I., Tatì, A., Romano, R., De Canio, G., 2018. Motion magnification analysis for structural monitoring of ancient constructions. *Measurement* 129, 375–380.
- Fleet, D.J., Jepson, A.D., 1990. Computation of component image velocity from local phase information. *Int. J. Comput. Vis.* 5, 77–104.
- Freeman, W.T., Adelson, E.H., 1991. The design and use of steerable filters. *IEEE Trans. Pattern Anal. Mach. Intell.* 13 (9), 891–906.
- Gautama, T., Van Hulle, M.A., 2002. A phase-based approach to the estimation of the optical flow field using spatial filtering. *IEEE Trans. Neural Netw.* 13 (5), 1127–1136.
- Kim, S.W., Yin, S., Yun, K., Choi, J.Y., 2014. Spatio-temporal weighting in local patches for direct estimation of camera motion in video stabilization. *Comput. Vis. Image Underst.* 118, 71–83.
- Lado-Roigé, R., Pérez, M.A., 2023. STB-VMM: Swin transformer based video motion magnification. *Knowl.-Based Syst.* 269, 110493.
- Liu, C., Torralba, A., Freeman, W.T., Durand, F., Adelson, E.H., 2005. Motion magnification. *ACM Trans. Graph.* 24 (3), 519–526.
- McLeod, A.J., Baxter, J.S.H., de Ribaupierre, S., Peters, T.M., 2014. Motion magnification for endoscopic surgery. In: *Medical Imaging 2014: Image-Guided Procedures, Robotic Interventions, and Modeling*. 9036, pp. 81–88.
- Ngo, A.C. Le, Johnston, A., Phan, R.C.-W., See, J., 2018. Micro-expression motion magnification: Global lagrangian vs. local eulerian approaches. In: *Proc. 2018 13th IEEE Int. Conf. Automatic Face & Gesture Recognition (FG 2018)*. pp. 650–656.
- Oh, T.-H., Jaroensri, R., Kim, C., Elgharib, M., Durand, F., Freeman, W.T., Matusik, W., 2018. Learning-based video motion magnification. In: *Proceedings of the European Conference on Computer Vision. ECCV*, pp. 633–648.
- Poozesh, P., Sarrafi, A., Mao, Z., Avitabile, P., Niezrecki, C., 2017. Feasibility of extracting operating shapes using phase-based motion magnification technique and stereo-photogrammetry. *J. Sound Vib.* 407, 350–366.
- Rother, C., Kolmogorov, V., Blake, A., 2004. GrabCut: Interactive foreground extraction using iterated graph cuts. *ACM Trans. Graph.* 23 (3), 309–314.

- Sarraf, A., Mao, Z., Niezrecki, C., Poozesh, P., 2018. Vibration-based damage detection in wind turbine blades using phase-based motion estimation and motion magnification. *J. Sound Vib.* 421, 300–318.
- Shabi, O., Natan, S., Kolel, A., Mukherjee, A., Tchaicheeyan, O., Wolfenson, H., Kiriyati, N., Lesman, A., 2020. Motion magnification analysis of microscopy videos of biological cells. *PLOS ONE* 15 (11), e0240127.
- Shao, H., Luo, L., Qian, J., Chen, S., Hu, C., Yang, J., 2023. Tranphys: Spatiotemporal masked transformer steered remote photoplethysmography estimation. *IEEE Trans. Circuits Syst. Video Technol.*
- Singh, J., Murala, S., Kosuru, G., 2023. Multi domain learning for motion magnification. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 13914–13923.
- Takeda, S., Akagi, Y., Okami, K., Isogai, M., Kimata, H., 2019. Video magnification in the wild using fractional anisotropy in temporal distribution. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 1614–1622.
- Takeda, S., Niwa, K., Isogawa, M., Shimizu, S., Okami, K., Aono, Y., 2022. Bilateral video magnification filter. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 17369–17378.
- Takeda, S., Okami, K., Mikami, D., Isogai, M., Kimata, H., 2018. Jerk-aware video acceleration magnification. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. pp. 1769–1777.
- Thomas, G., Gade, R., Moeslund, T.B., Carr, P., Hilton, A., 2017. Computer vision for sports: Current applications and research topics. *Comput. Vis. Image Underst.* 159, 3–18.
- Wadhwa, N., Rubinstein, M., Durand, F., Freeman, W.T., 2013. Phase-based video motion processing. *ACM Trans. Graph.* 32 (4), 1–10.
- Wadhwa, N., Rubinstein, M., Durand, F., Freeman, W.T., 2014. Riesz pyramids for fast phase-based video magnification. In: *2014 IEEE International Conference on Computational Photography. ICCP*, pp. 1–10.
- Wang, Y., See, J., Oh, Y.-H., Phan, R.C.-W., Rahulamathavan, Y., Ling, H.-C., Tan, S.-W., Li, X., 2017. Effective recognition of facial micro-expressions with video motion magnification. *Multimedia Tools Appl.* 76, 21665–21690.
- Wang, F., Zhong, B., Liu, T., Kong, X., Bai, H., Wan, G., 2024. Deep matching online video stabilization using TSNet. *IEEE Trans. Instrum. Meas.*
- Wang, X., Zhu, Z., 2023. Context understanding in computer vision: A survey. *Comput. Vis. Image Underst.* 229, 103646.
- Wu, H.-Y., Rubinstein, M., Shih, E., Guttag, J., Durand, F., Freeman, W., 2012. Eulerian video magnification for revealing subtle changes in the world. *ACM Trans. Graph.* 31 (4), 1–8.
- Wu, X., Yang, X., Jin, J., Yang, Z., 2018. PCA-based magnification method for revealing small signals in video. *Signal, Image Video Process.* 12 (7), 1293–1299.
- Yang, Y., Dorn, C., Mancini, T., Talken, Z., Kenyon, G., Farrar, C., Mascareñas, D., 2017. Blind identification of full-field vibration modes from video measurements with phase-based video motion magnification. *Mech. Syst. Signal Process.* 85, 567–590.
- Zhang, Y., Pinteá, S.L., van Gemert, J.C., 2017. Video acceleration magnification. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. pp. 529–537.
- Zhang, L., Xu, Q.-K., Huang, H., 2015. A global approach to fast video stabilization. *IEEE Trans. Circuits Syst. Video Technol.* 27 (2), 225–235.
- Zhao, C., Wang, H., Chen, H., Shi, W., Feng, Y., 2022. JAMSNet: A remote pulse extraction network based on joint attention and multi-scale fusion. *IEEE Trans. Circuits Syst. Video Technol.* 33 (6), 2783–2797.